# MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) MODEL QUESTION PAPER

Semester V

## MG5DSEBCA302 - Machine Learning

(*2024 ADMISSION ONWARDS*)

**Duration: 2 Hours** **Maximum Marks: 70**

***Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)***

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

### Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks.

1. Define Machine Learning. How does it differ from conventional programming? [U] [CO1]
2. Distinguish between a labelled dataset and an unlabelled dataset. [U] [CO1]
3. Differentiate between regression and classification problems. [An] [CO2]
4. Analyse the significance of linear separability in the working of a Linear Perceptron. [An] [CO3]
5. Mention any two ethical considerations in Machine Learning. [U] [CO4]

**[2*5=10 Marks]**

### Part B

Short Answer Questions

Answer any 5 questions

Each question carries **6** marks.

6. Explain the Confusion Matrix. [U] [CO1]
7. Compare Simple Linear Regression, Multiple Linear Regression. [U] [CO2]
8. Analyse the structure and working of a Multilayer Perceptron in comparison with a Linear Perceptron. [An] [CO3]
9. Apply the K-Means clustering algorithm to group a given set of data points into an appropriate number of clusters. Explain the steps involved. [A] [CO4]
10. A large dataset requires efficient and scalable clustering. Explain how BIRCH can be applied for clustering such data. [U] [CO4]

11. Explain any three realworld applications of Machine Learning [U] [CO4]
12. Explain different clustering validation measures to assess the quality of the clusters obtained from a clustering algorithm. [U] [CO4]

**[6*5=30 Marks]**

## Part C

Essay Questions
Answer any 2 questions
Each question carries **15** marks

13. Describe the Machine Learning learning framework. [U] [CO1]
14. Analyse the importance of data preprocessing, feature selection, and feature extraction in developing Machine Learning models. [An] [CO2]
15. Analyse the architecture and working of a Feed Forward Neural Network. [An] [CO3]

**[15*2=30 Marks]**